Balanced assignments of periodic tasks

Work with Frédéric Meunier (CERMICS)

Héloïse Gachet

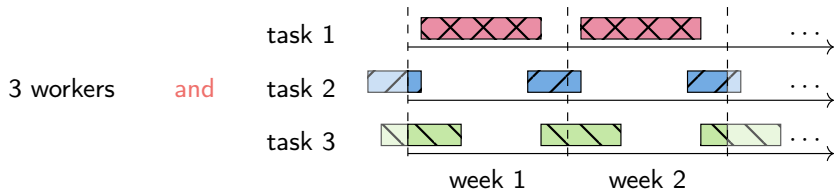
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Introduction

- Tasks $[n]$ to be performed at the same time every week
- Number q of indistinguishable workers



Aim. Find a feasible assignment of tasks to workers so that each worker performs each task with same asymptotic frequency
$\rightarrow$ Balanced assignment

Assignment: $f : [n] \times \mathbb{Z}_{>0} \to [q]$, where $f(i, r) = j$ means worker j performs task i on week r

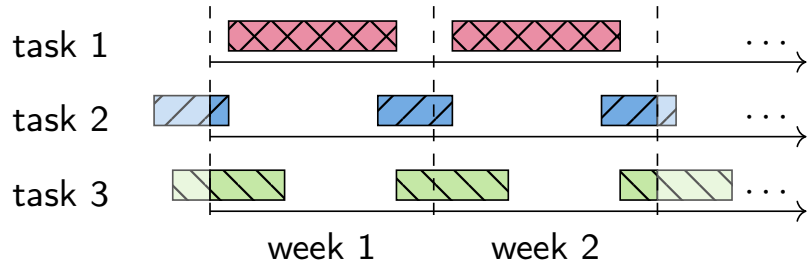
Feasible assignment: f is non overlapping.

Balanced assignment:

$$f \text{ feasible} \quad \text{and} \quad \lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i, r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

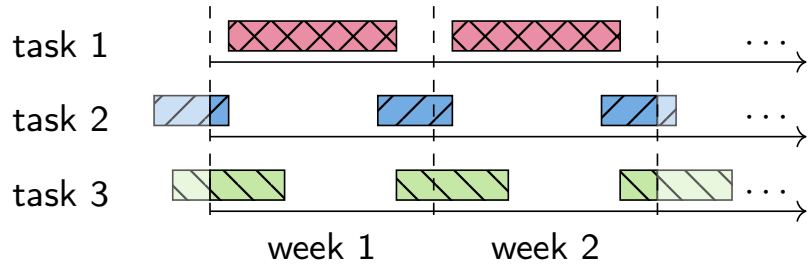
Feasible and balanced assignments: examples

3 workers and

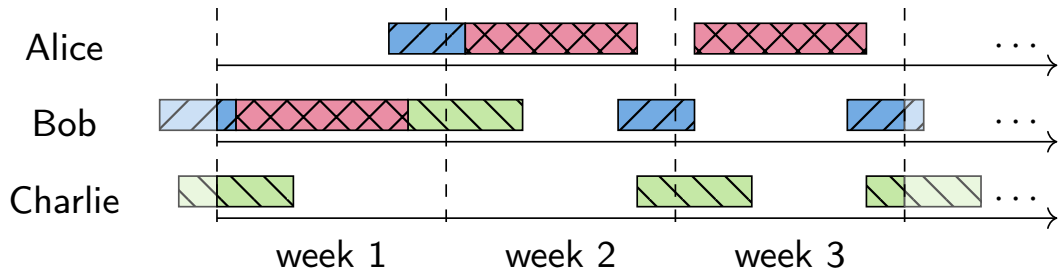


Feasible and balanced assignments: examples

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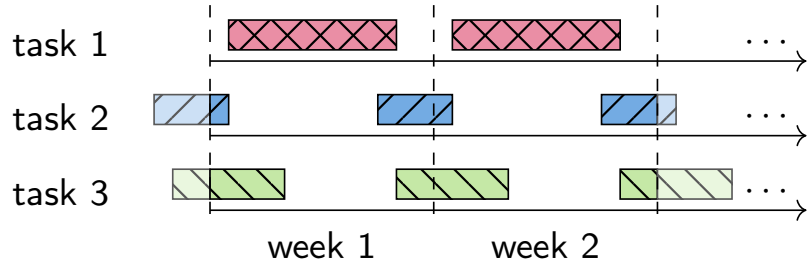


Feasible assignment:

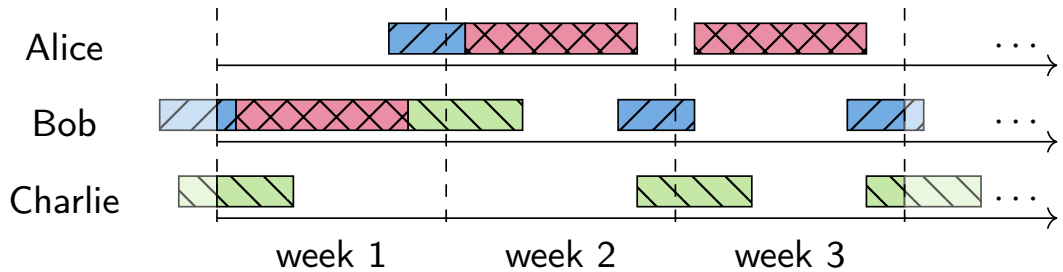


Feasible and balanced assignments: examples

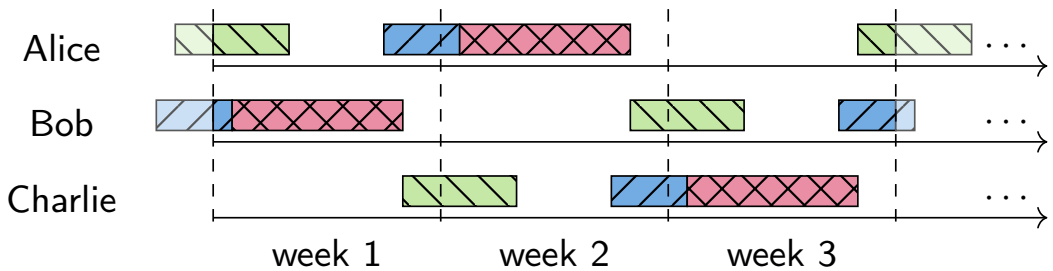
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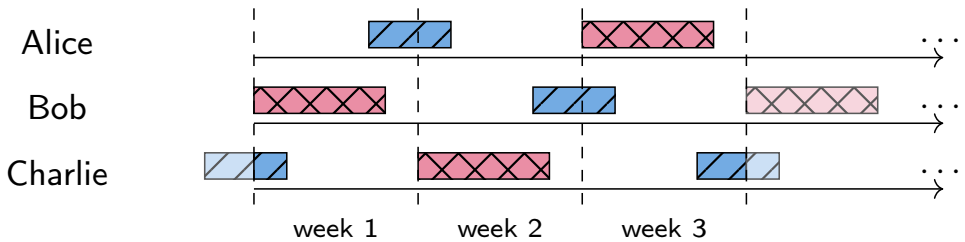
Feasible assignment:



Balanced assignment:



Main results



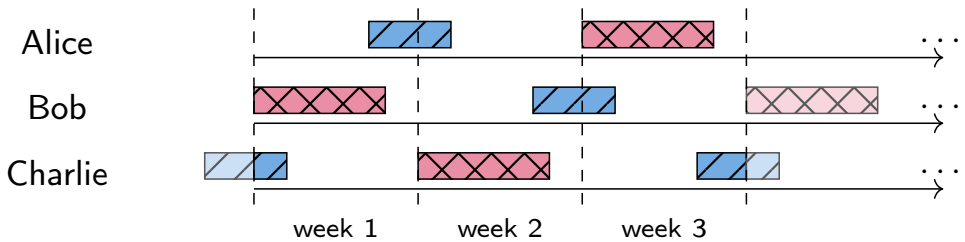
$$\lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i,r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

There exists a balanced assignment.

$\Downarrow$

There exists a feasible assignment with a worker performing each task at least once.

Main results



$$\lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i,r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

There exists a balanced assignment.

⇳

There exists a feasible assignment with a worker performing each task at least once.

Main results

Theorem G. , Meunier

The following statements are equivalent:

1. There exists a feasible assignment with a worker performing each task at least once.
2. There exists a balanced assignment.
3. There exists a periodic balanced assignment with period q.

q: number of workers

Algorithmic proof with polynomiality results:

- Deciding existence of a balanced assignment
- Computing a periodic balanced assignment with period q

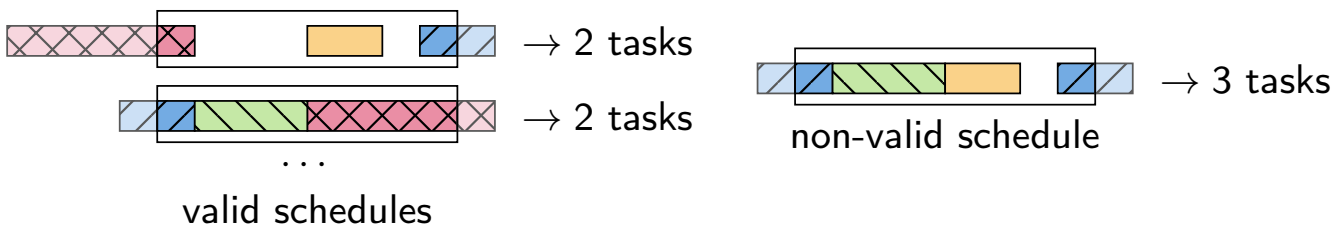
Extended version: setting

Input. n tasks to be repeated every week, q workers, set $\mathcal{S}$ of valid weekly schedules

Feasible assignment: f non overlapping and induces schedules in $\mathcal{S}$

Examples:

- workers perform at most M tasks per week.
  with $M = 2$



- workers work at most H hours per week.
- when q tasks have to be performed at time zero, workers cannot teleport.
- ...

Balanced assignments: results

Theorem G., Meunier

If there exists a feasible assignment with a worker performing each task at least once and q tasks have to be performed at time zero, then there exists a balanced assignment.

q: number of workers

Examples where the additionnal condition is needed

Theorem G., Meunier

If there exists a balanced assignment, then there exists a periodic balanced assignment with period bounded by $2(q-1)q!$.

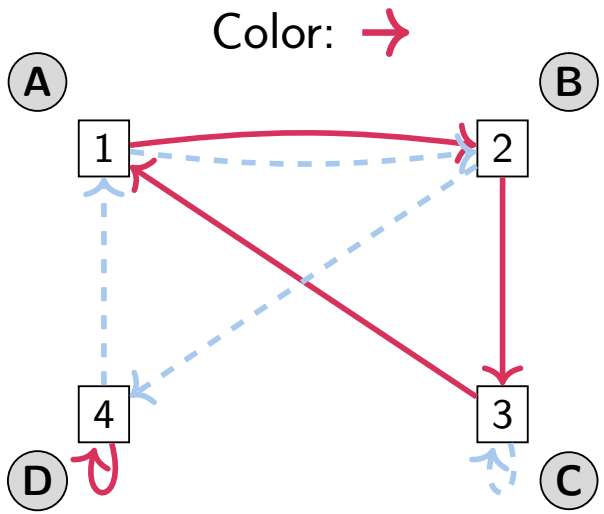
- Existence of a balanced assignment decidable

A problem of pebbles on a graph

Consider:

- graph $D = (V, A)$ with colored arcs, each color being a permutation over V
- one pebble on each vertex

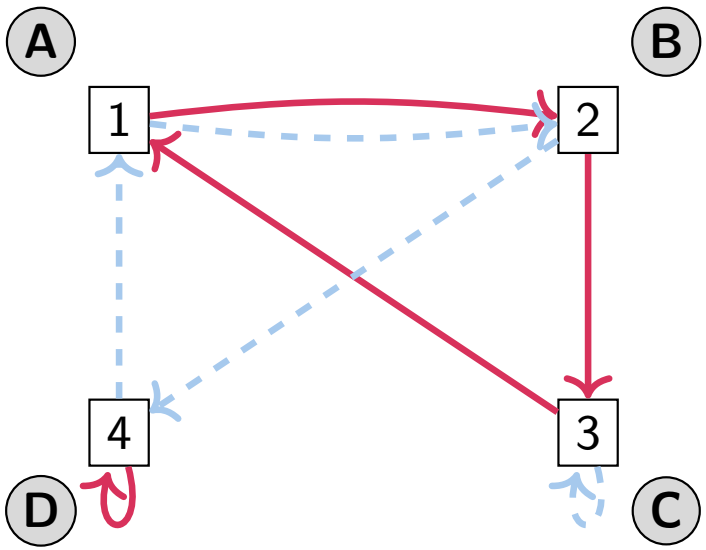
Sequence of colors defines sequence of moves of the pebbles



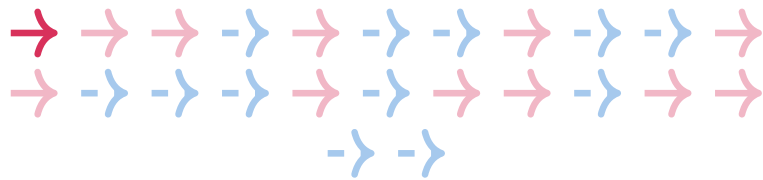
- Vertex = task at time zero
- Arc = valid schedule
- Pebble = worker
- Color c = partition of the tasks into schedules

Balanced sequence of colors: example

Aim. Find a sequence of colors making each pebble visit each arc with same asymptotic frequency.



Sequence of colors:



Arcs visiting:

	1→2	2→3	3→1	4→4	1-›2	2-›4	3-›3	4-›1
A	**1**	0	0	0	0	0	0	0
B	0	**1**	0	0	0	0	0	0
C	0	0	**1**	0	0	0	0	0
D	0	0	0	**1**	0	0	0	0

Balanced sequence of colors: result

Proposition G., Meunier

If D is connected, then there exists a sequence of colors making each pebble visit each arc with same asymptotic frequency.

Two proofs:

- Markov chains: non constructive
- Eulerian cycle in the Cayley graph associated to the set Σ of colors (seen as permutations of V):
  - Contructive proof
  - Additionnal result: the sequence of colors can be made periodic (period upper bounded by $|A| \times |\langle\Sigma\rangle|/|V|$)

Balanced assignments: graph connected with $2(q-1)$ colors
$\rightarrow$ period upper bounded by $2(q-1)q!$
q: number of workers

Perspectives and open questions

Question

Is the bound $2(q-1)q!$ for periodic balanced assignments tight for a single instance?

Partial answer:

Corollary G., Meunier

If q is a prime and there exists a balanced assignment, then there exists one that is periodic with period upper bounded by $q(q!-1)$.

q: number of workers

Using our graph framework along with:

- Group theory tools: Cauchy theorem, orbit-stabilizer theorem

Thank you!